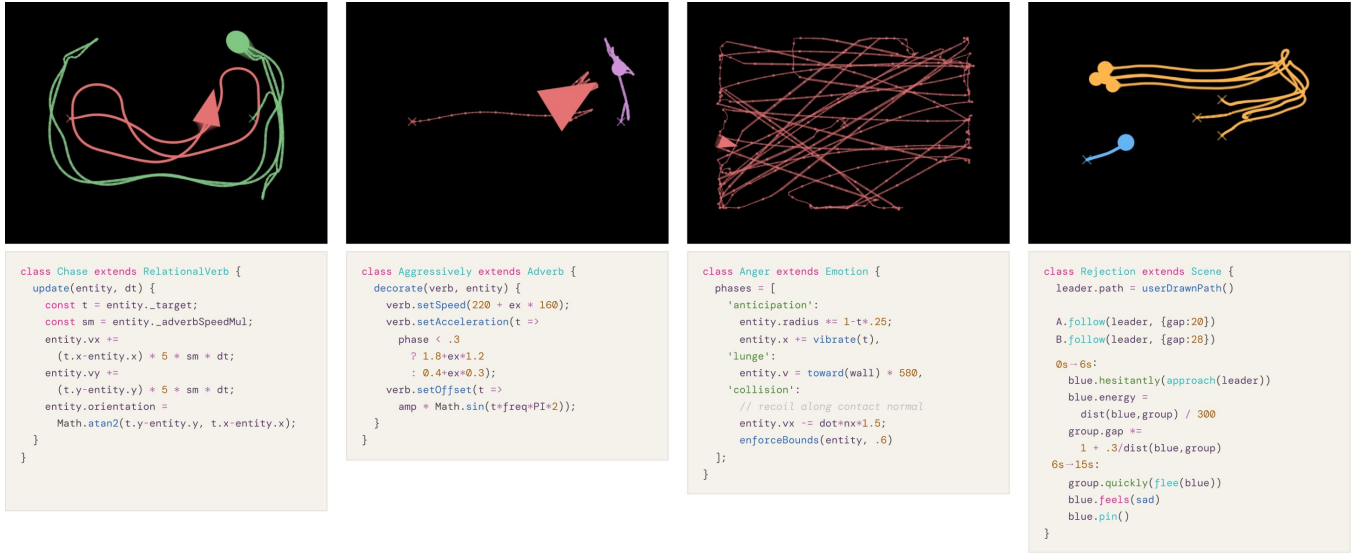


# fog: Expressing Emotions and Social Dynamics through Function Composition of AI-Generated Code

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**Figure 1:** In *fog*, users utilize LLMs to generate motion and emotion functions by implementing class definitions for verbs, adverbs, emotions, and gestures. These functions can compose with each other and act over an entity’s internal and external state (energy, velocity, collision, paths) to show behavior. The first three gallery examples show instances of social motion (chase), motion decoration (aggression), and emotionally-expressive motion (anger). The last example demonstrates how a function composition can demonstrate social dynamics like loneliness and belonging.

## Abstract

Motion and emotion are core parts of intelligent, expressive behavior. In this paper, we introduce *fog*, a function composition framework for implementing and compose motion functions. We demonstrate how *fog* can be used to express social dynamics and emotions in Heider-Simmel-style animations. This code generation framework can help users generate functions for verbs, adverbs, gestures, and emotions to create an open-ended motion vocabulary. It is complemented by an animation editor that helps users refine motion through direct manipulation and dynamically generated UI. We evaluate our approach with a perceptual evaluation, where we test 452 *fog*-generated animations to see if people can recognize

the semantic meaning of the motion. We find that *fog*’s motion functions can be recognized at 68% accuracy, a 2.68x improvement over a chance baseline. In a mixed-methods user study with professionals and novices, we show that *fog* in interface form can support users with more rapid iteration, exploration, and control.

## CCS Concepts

• **Human-centered computing** → *Natural language interfaces; User interface programming*; • **Applied computing** → **Media arts**; Psychology.

## Keywords

motion, emotion, animation, function composition, generative AI, code generation, human-AI interaction, Heider-Simmel, generative user interfaces

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# 1 Introduction

Motion is emotion without the ‘e’. As people, we tend to interpret meaning, intentionality, and animacy from things that move. This has been well established in cognitive science. In 1944, Heider and Simmel [12] ran a famous psychology experiment using an animation where three shapes played out a series of events. They found that even when the animation ran without sound or words, participants still described a story of love, joy, fight, and chase. This experiment showed that motion alone can narratively express emotions and social dynamics.

However, it is hard to animate emotion and social dynamics. Even simple moments are hard to get right. How do you make something look happy or sad? How do you represent conflict between two characters? These tensions are the fundamental challenges of storytelling, and this challenge of animating emotion and social dynamics through motion is the problem we address in this paper. Animation has previously explored this problem through the lens of character animation, puppet animation, physics-based animation, and so on. In this paper, we choose to look at Heider-Simmel animations, where stories are acted out by shapes. While these animations are simple and abstract in motion representation, they can help us understand what building blocks are necessary for spatial behavior to convey story.

Today, AI can create visual storytelling by generating video and animation. However the challenges of control with AI that have been extensively discussed in text and image only get harder with animation, where motion is dynamic across both space and time. For example, if a character is moving through a scene, it becomes hard for the user to describe exactly how fast or slow the character is going, what path they should take to get from A to B, and whether they want the motion to look hesitant, urgent, or soft. AI tools do not give users ways to easily explore and iterate on motion dynamics such as speed, force, easing, and timing.

Our goal in this paper is to bridge the symbolic nature of storytelling with a numerical understanding of motion and to give people control with AI over both aspects. To do so, we draw upon theories from computer graphics [22] and cognitive psychology [26] to create a hierarchy of functions that express behavior. Our main insight is that there is a connection between motion and emotion – both reduce down to forces, energy, and states in time. We can capture the semantic dimensions of both motion and emotion in class-based abstractions, so AI can implement new instances and help users create an open-ended vocabulary of animation functions.

We present *fog*, a function-composition framework that demonstrates how AI can implement Motion and Emotion classes to produce animation functions spanning Verbs, Adverbs, Gestures, and Emotions. In this framework, Motion is applied onto entities that accept function compositions manipulating their external and internal state. For example, entities can chase and avoid each other, instancing Verbs to create attraction and rejection in motion. Adverbs and Emotions can compose with these verbs to modify motion performance. A child entity can hesitantly(chase) a guardian entity with a stop-and-go motion and nervous look-arounds (through velocity and squash-and-stretch changes). An entity can desperately(avoid) a predator by making sharp and

wild turns as it flees. Function composition is achieved by having abstract classes such as Verbs and Adverbs contract on channels like speed, acceleration, and easing. *fog* also supports Gestures, which add little units of socially meaningful motion on top of primary motion, like waving or nodding. Lastly, *fog* supports the expression of Emotion through motion. For example, Anger can be expressed in phases of motion as tremors (a build-up of internal turmoil), an aggressive lunge towards a wall (expression towards a target), and collision (recoil). This sequence of affective motion is depicted in visual and code form in Fig. 1.

*fog*’s function-composition architecture is also made for freedom and control in user interaction. Users can generate new motion functions, draw paths to bind to functions, and dynamically generate UI to tune and tweak function compositions on axes like speed, intensity, and exaggeration. To evaluate *fog*, we run a perceptual evaluation that benchmarks the range and recognizability of its automatically generated motion. We also conduct a user study to see how professionals and novices can use the system to interactively animate Heider-Simmel-style stories. We conclude with a discussion of how abstract classes like *fog* can be new surface area for prompts and how users can define their own layers of abstraction between story, physics, motion, and emotion.

Our contributions are the following:

- *fog*, a function composition framework of abstract classes for motion and emotion. It introduces a generative vocabulary of functions spanning Verbs, Adverbs, Gestures, and Emotions.
- An animation editor that shows how users can 1) generate new motion functions on-the-fly, 2) compose functions into scenes and stories, and 3) use a suite of interaction techniques from prompts to direct manipulation controls for motion refinement.
- Perceptual evaluation testing 452 animations (spanning 32 verbs, 12 adverbs, 12 gestures, 12 emotions and their function compositions) to see if people can recognize the semantic meaning of *fog*-generated motion. We demonstrate that people can match the motion to the verb, adverb, gesture, or emotion at 68% accuracy, a 2.68x improvement over a chance baseline.
- Mixed-methods user study (n=10) with 6 professionals and 4 novices showing that *fog* enables more rapid iteration compared to a prompt-based baseline and provides support for user control and exploration.

## 2 Related Work

### 2.1 Theories and Early Work in Animation

Our ability to interpret animacy and intent from motion is the core result of the Heider-Simmel paper [12]. In this experiment, participants described how in an animation of only simple shapes, they could see a story play out with emotions and social dynamics: the bigger triangle, *T*, was bullying the smaller shapes, *t* and *s*. Heider-Simmel interpreted these results as evidence that people can understand causal origins of motion and interpret social intent even with abstract characters.

Other results from cognitive psychology have shown that from a young age, we know the difference between animate and inanimate

motion [9, 26]. Animate motion is motion attributed to living beings, while inanimate motion is driven by physics acting on objects (e.g. billiard balls). The distinction is important for animation systems focused on expressing stories, because systems that focus too much on getting the physics right lose a life-like quality to motion. In fact, it is often the departure from expected physics in animation that gives it the "illusion of life" [15]. This is often described in the twelve fundamental principles of animation outlined by Disney animators Frank Thomas and Ollie Johnston. Character animation works well when there is exaggeration, squash-and-stretch, compositions of primary and secondary motion, arcs, and so on.

Early ideas in how to represent the motion of autonomous, intelligent, and expressive animation came from computer animation in the 1980's. Reynolds in his influential paper on steering behaviors [22] described how human behavior can be considered a "hierarchy of motion behaviors" that spanned action selection, steering, and locomotion implementation. He demonstrated that by composing velocity and acceleration forces, entities could be animated to flee and seek towards targets or scenarios like obstacle navigation. This work was primarily physics-driven. Early work also focused on computer animation for emotion and expressive performance [2, 17]. Strassman's "Desktop Theater" system [25] converted natural language into animation by storing a library of motion behaviors and keeping both a numerical and symbolic representation of animation. This allowed Divaldo to animate scenes with different sets of motion dynamics like anger and rudeness.

## 2.2 HCI Systems for Animation

A large body of HCI work explores how animation can benefit from inputs and controls like sketch, borrowed dynamics, physics, templates, blocks and presets [5, 14], and design principles [20]. Systems like Draco [18] EnergyBrushes [27] layered motion on top of animation using path and particle motion as well as vector fields. Tools [16, 28] utilized motion dynamics from video for motion re-targeting. Other systems have explored the combinations of inputs such as sketch with voice [23] and performance (gesture) with graphic overlays [24], to leverage the complementary strengths of different modalities.

## 2.3 Generative AI for Motion

AI systems prior to generative AI have looked at emotion-driven motion. Geppetto [8] was a system that helped users create emotionally expressive robot motion – designing a robot to move in a way that was happy versus sad. One of the first approaches to generative animation was Spellburst [1], which demonstrated that LLMs can generate code and enable new interactions like code blending. Other ensuing systems like Keyframer and LogoMotion demonstrated that AI can enable novices to animate by building rich intermediate representations of the animation state and giving users high-level controls. While the first systems were largely text- and prompt-driven, systems now have introduced new ways of binding natural language to GUI interactions. For example, Narrative Motion Blocks [4] demonstrate how sketch input and paths could combine with prompts to direct path-driven motion. Toyteller [7] also looked at the problem of Heider-Simmel animations. They trained a transformer that learned a representational space between

natural language and motion. Other machine learning approaches have demonstrated that we can introduce motion prompts [10] as a way of representing desired intent for motion without natural language. Goel et. al. [11] demonstrated that it was possible to train diffusion models to learn motion edits for the purposes of facilitating iteration. Motion controls and editing interactions actively being developed ([3], motion modules [13], image- and frame-based controls [6, 19], and motion prompts [10]. *fog* builds on this body of work by contributing a framework to semantically represent motion, dynamics, and emotion in functions that connect to user input and interaction.

## 3 Formative Work

Animating emotions with only simple shapes in the style of Heider-Simmel animations is not the most common animation problem. To understand how to approach it better, we conducted a few formative steps. First, we talked to professional animators to understand what their approach would be, what tools they use today, and how their workflows are evolving with AI.

### 3.1 Interviews with Professional Animators

**3.1.1 Methodology.** We conducted three semi-structured interviews with animators. We reached out to animators on Upwork through purposive sampling. We interviewed three professionals (E1-E3) who were chosen based on their extensive portfolios of motion graphics and animation work. E1 had experience in After Effects, Cinema4D, as well as AI design tools. E2 was a designer who primarily worked on logos and did animations upon client request. They had experience with After Effects and some exposure to other animation tools like Rive, Figma, and Framer for animation. E3 was a character animator who worked primarily in Toon Boom Harmony. The interviews were semi-structured: first, we had professionals walk through one or two projects in their portfolio to talk through their workflow and tools; then, we asked them how they would approach the animation of a simple shape to express emotion. In total, interviews lasted approximately 1-1.5 hrs, and professionals were compensated \$40 for their time and expertise.

**3.1.2 Findings.** First, animators were experimenting with AI tools through custom scripts. E1 and E3 had all experimented with tools like ChatGPT and Claude to create custom programs that would help them with friction points in their workflow. For example, E3 had asked Claude to create a script that could fill in inbetween frames for them. E1 took scripts from ChatGPT and pasted them in as custom timeline expressions in After Effects to show how they could get the system to support rigidbody physics where it could not before.

Second, animators often searched for professional presets. Even when animators are professionals, they often have to reach for presets. E2 described how most of their recent work for clients was focused on creating a polished and premium motion feel. To get that, they often used components shared in communities. While these components worked great for E2 in isolation, they could not be easily combined – they lacked the shared state and tool awareness to support composition.

Third, during the exploratory task, we found that animators made use of everything from paths, speed, energy, and deformation

to animate emotion on a simple shape. When asked to animate sadness, E3 sketched out a shape and explained how they would use exaggeration and a droopy deform to make a circle look sad, “It needs a sort of exaggeration in the opposite direction of the movement before it starts moving or else it looks robotic. So I always do a slight exaggeration since it’s going to droop downwards and flatten.” They would keyframe and animate using a combination of squash-and-stretch, exaggeration, and anticipation. They also mentioned that while they expressed sadness in one way (with deformation), they could express another emotion like anger with completely different channels. “Anger is a stiffer emotion. It can rattle around and be flying around the screen to look angry.”

These interviews motivated our final set of goals for *fog*:

- **Provide people with a basic motion vocabulary, but make it an open set so users can dynamically generate new primitives.** The *fog* framework should intuitively match the vocabulary people have for motion, such as verbs adverbs, and gestures.
- **Enable function composition and blending across primitives.** Animators mentioned that presets are powerful but difficult to compose. *fog* should help users quickly create complex compositions. These presets should be able to share and meaningfully manipulate each other’s state.
- **Give users a spectrum of direct manipulation and language controls built over what functions compose on.** Animators noted that tweaking exaggeration and speed are key for motion refinement. *fog* should provide users with controls for perceptual, hard-to-describe things like paths and speed and connect them with prompt-based interaction

## 4 System

### 4.1 The *fog* Framework

*fog* is a hierarchy of abstract classes for expressing motion and emotion. These abstract classes include Scene, State, Entity, Emotion, Motion, Verb, Adverb, Gesture, and Emotion. They scaffold code generation and bridge the higher-level symbolic aspects of motion with its lower-level numerical underpinnings (components like velocity, acceleration, intensity, and exaggeration).

**4.1.1 Scenes, States, Entities.** Scenes are our high-level representation of the animation. They consist of a series of States that describe the animation on entities in terms of primary motion and secondary motion. The Entity class represents characters as shapes. It keeps track of position, velocity, energy, position history, and orientation. Each entity has the following visual properties (shape: {circle, triangle}, color, radius). To give entities a sense of animacy, we apply squash-and-stretch to stretch entities in the direction of their velocity, and give them motion trails for tails. Each entity accumulates acceleration into velocity and integrates velocity into position: `entity.vy += this._gravity * dt;` `entity.x += entity.vx * dt.`

**4.1.2 Verbs.** Verbs extend the Motion class and were a type of primary motion. Verbs could be Solo, Relational, PathMotion, or PinMotion verbs. Solo verbs were actions like struggle, walk, and idle. Relational verbs were actions that occur in between characters such as follow, chase, and avoid. Solo verbs had helper

force functions like impulse and spring, and Relational verbs had helper force functions like personalBubble, lookAt, attract, repel, collision, mirror, and match. PinMotion allows a verb to happen in-place. PathMotion allows entities to follow along Paths. Paths can be generated programmatically (a figure-eight orbit) or take in user-drawn paths. This allows *fog* to flexibly support both complex trajectories (arcs, spirals, polylines) as well as freehand sketches.

**4.1.3 Adverbs.** Adverbs are a class that extend SecondaryMotion. They decorate PrimaryMotion. Adverb(Verb) is one of the most intuitive and clear function compositions in *fog*, since it parallels natural language. The abstract classes contract with each other on channels such as: speed, acceleration, exaggeration, easing, positional offsets, pauses, and time windows. Adverbs can deterministically set motion dynamics, and Verbs to read and respect it, because they share state. Entities can be made to move Slowly (with low pixels/second) or Quickly (with high pixels/second, like 512 px/s, crossing the canvas in 1 second). However, function composition goes beyond simple get and set; complex formulas can be generated to create rich dynamics and motion envelopes. For example, for `aggressively(chase(B))`, the adverb aggressively can provide a sawtooth-shaped acceleration function, sharp jitter from positional offsets, and high speed to compose on top of the chase verb’s own velocity writes and intent-driven motion.

**Table 1: Function Compositions in *fog* Animation Framework**

| Function Compositions    | Example                                     |
|--------------------------|---------------------------------------------|
| Relational Verb          | <code>follow(B)</code>                      |
| Adverb + Verb            | <code>hesitantly(pace)</code>               |
| Adverb + Relational Verb | <code>hesitantly(follow(B))</code>          |
| PathVerb + Path          | <code>zigzag(Path), follow(drawPath)</code> |
| Gesture + Verb           | <code>wave + walk</code>                    |
| Adverb + Gesture         | <code>slowly(clap)</code>                   |
| Emotion + Verb           | <code>angrily(chase(B))</code>              |
| Emotion + Emotion        | <code>anger + fear</code>                   |
| Emotion + Emotion + Verb | <code>(anger + fear) charge</code>          |

**4.1.4 Gestures.** Gestures are SecondaryMotion that decorate instances of PrimaryMotion with small units of semantically meaningful motion. For example, wave and nod are two gestures *fog* implements. A wave can be implemented by spawning a particle for a hand and rotating it within close range. A nod or the headShake can be implemented by sinusoidally squashing-and-stretching along the vertical or horizontal axis a few times. Each gesture implements a shapeFunction helper method, which maps the temporal dynamics of the gesture across progress through the state. For example, `clap.shapeFunction()` graphed out would show the “attack-hold-release” pattern inherent in the motion. A few shapeFunction graphs are illustrated in the Supplementary Material.

Gestures can apply continuously throughout the state or cycle a few times up to a count (e.g. entity nods twice). Gestures can also take up a local window of time within a state. For example, an entity can wave once just before walking over or continuously

wave as they walk. Gestures can also function compose with adverbs and verbs. Adverb(Gestures) like `slowly(clap)` can decorate gestures along the same channels mentioned for verbs (acceleration, easing). Verb + Gestures also compose naturally as primary motion paired with secondary motion. Sometimes function composition can have multiple functions targeting the same channels. For example, if an entity is wandering about but also nodding, its velocity and squash-and-stretch can be written to both by both functions. In these cases, the effects are multiplicative or additive depending on the functions combined.

**4.1.5 Emotions.** Lastly, Emotions are a class for affective motion. Emotions hold Phases to create an event-driven sequence of motion. For example, Anger could be expressed through three phases: anticipation (vibration motion for build-up), lunge (driving towards a wall target), and collision (boundary behavior enforcement, recoil along the contact normal, and a drop in velocity from energy loss). Emotions implement: 1) `energyFunction` and 2) `customDraw`. The `energyFunction` modulates the motion performance based on the entity's energy budget. For example, if the emotion is Sadness, the entity can lose energy until it is just making slight, lethargic movements in place. If the emotion is Ecstatic, the energy can ramp and go up with impulses each time the entity bounces around the canvas. The `customDraw` method implements particle effects bespoke to the emotion. For example, another way to express Sadness, is to spawn particles that look like tears and crying.

Emotions can stand on their own or compose with other classes. `angrily(follow)` composes Emotion with Verb. To allow for this flexibility, each Emotion also implements a `decorate` method (like Adverbs) to provide a motion signature characteristic to the emotion. For example, to represent Fear, the `decorate` method can always compose a tremor on the entity's primary motion. Emotions can also compose with each other to produce a *blend* of emotions (anger+fear). *fog* supports three different types of function compositions for blending: *interpolate*, *interleave*, and *blend by dimension*. Interpolation is supported by capturing the delta from each emotion and linearly interpolating. Interleaving is supported by interspersing Phases from each of emotion. Lastly, *blend by dimension* allows channels to be picked from each emotion for composition (e.g. take `energyFunction` from A, but take speed from B).

**4.1.6 fog User Interface.** The point of *fog* as a framework was not only to define a ladder of class abstractions going from high-level and semantic to numerical and force-driven, but also to create openings in the abstractions that lend well to user interaction and direct manipulation in an graphical user interface. Fig. 2 shows what *fog* looks as an animation editor for the user. It consists of a Main Canvas, a Timeline, an State Editor. The Canvas was where the animation (Fig. 2-A) played out. Underneath the canvas, users could pass in prompts (Fig. 2-B). The Timeline (Fig. 2-C) helped visualize the Scene and its States and gave users playback controls. Clicking on a state opened up an isolation mode for detailed editing. At the top of each state, a state description described the function compositions in that state in a grammar form close to natural language. Underneath each entity, a Scene Table (Fig. 2-E) showed the motion functions on each entity and gave users an entrypoint to explore (Verb, Adverb, Gesture, Emotion) functions. For example, the Verb grid in (Fig. 2-F) let users explore different

Solo, Path, Relational verbs that came as pre-generated presets. If the user wanted to a new behavior, they could generate their own by writing a prompt and *fog* would infer which class to inherit and implement. If they needed ideas, a suggest button helped them come up with motion functions relevant to their state and scene. These generation functions show under (Fig. 2-H). This function generation pattern was also the same for Adverbs, Gestures, and Emotions. One feature specific to Verbs was that when a `PathVerb` was active, users could draw paths using the Pencil (Fig. 2-G), binding user-drawn paths to an active `PathVerb` (Fig. 3-B).

Next, users could advance into Adverbs to change speed and performance of the entity motion. They could try pre-generated adverbs (Slowly, Quickly, Erratically) or generate their own (similar to the Verb Panel), but they could also use the XY minimap for Speed and Intensity by clicking and dragging around. We show the effects using this minimap on motion dynamics in Fig. 3-A and Fig. 3-B; setting the intensity and speed high produced more force and velocity (shown by the sparse spacing across time), setting speed and intensity to be low produced more dense and slow movement across time. Speed and intensity were parameters that were baked into class implementation during generation time, but other tunable parameters also baked into different classes could also be axes. For example, Emotion opens a minimap for energy and exaggeration.

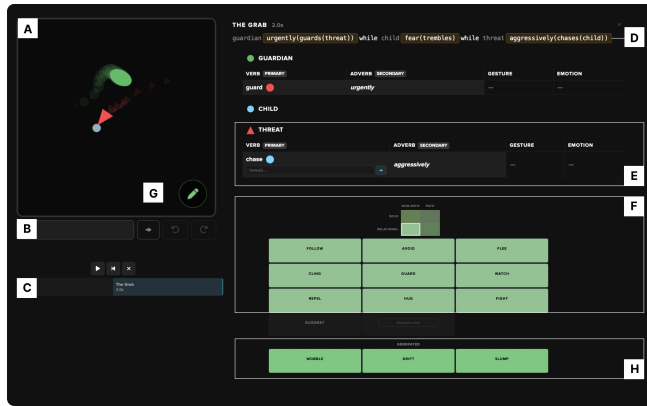
When users were working on a function composition, they could open up a refinement panel to have AI refine and rewrite the implementation of each class. This helped adapt the composition so that it became aware of context from the scene and state (aggressivelyFollow would know it exists in the Grab state of a Scene with a guardian and child entity). The refinement panel is shown in Fig. 3-E-H. It shows the channels both functions act on. Beneath the channels, an XY minimap would generate on demand, surfacing axes bespoke to the refined composition. For example, for aggressivelyFollow, Amplitude could be on the x-Axis, and Aggression on the y-Axis. Users could also use prompts to tweak the refined function. This pattern for Adverb(Verbs) was the same for Emotion(Verb) and Emotion+Emotion. More System Implementation details can be found in the Supplementary. Overall, the point of having this global-to-local drilldown, from global prompts to encapsulated functions and direct manipulation refinement, was to help users get instant feedback, deterministic steps, and suggestions for iteration.

## 5 Evaluations

We conducted two evaluations to understand *fog*'s capabilities: 1) a perceptual evaluation with crowdworkers to judge the recognizability of automatically generated motion functions and benchmark its expressive range on 452 animations, and 2) a user study (n=10) to understand how *fog*'s interaction paradigm would take with professionals and novices for animation editing.

### 5.1 Motion Recognition Perceptual Evaluation

Our perceptual evaluation was to answer the following two questions. First, RQ1) To what extent could these abstract classes for Adverbs, Verbs, Gesture, and Emotion generate a vocabulary of

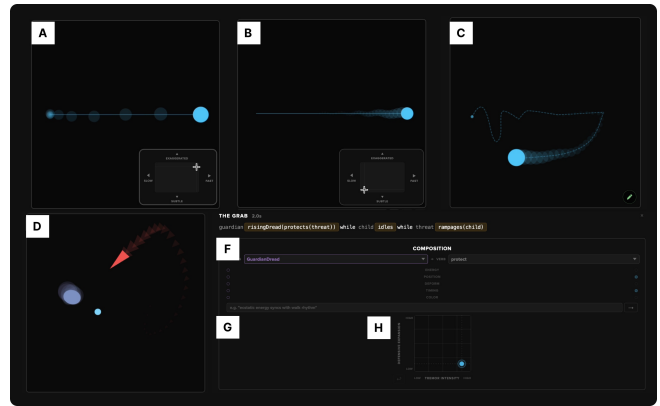


**Figure 2: fog as an animation editor with timeline, scene, and state support.** Users can edit by composing motion functions and generating functions on-the-fly using fog’s abstract classes for verbs, adverbs, gestures, and emotions.

motion primitives? Second, RQ2) To what extent could these motion functions compose and still produce semantically meaningful motion? Our hypotheses were that: H1) The system could produce relevant and diverse motion signatures across each category, but only up to a coarse degree of motion resolution. The closer words in each dimension got in terms of semantic meaning, the closer and more confusable the motion signatures were with each other, because this system has limited ways to animate simple shapes. H2) The motion functions would capably compose, but there would be certain cases where the additive nature of composition would cancel contributions.

**5.1.1 Perceptual Evaluation Methodology.** We conducted our perceptual evaluation study by having crowdworkers judge animations using a four alternative forced-choice task setup. First, we evaluated each of the classes in isolation to understand how recognizable the automatically implemented functions were. Then we evaluated the abstract classes in composition (Adverb(Verb), Verb+Gesture, Emotion(Verb)). For each motion dimension, we chose a set of 12 stimulus words to balance for variety and coverage. For adverbs, we chose 12 split across speed and manner {*slowly*, *quickly*, *suddenly*, *urgently*, *eagerly*, *forcefully*, *hesitantly*, *nervously*, *lazily*, *erratically*, *gently*, *playfully*}. For verbs, we chose 32 words balanced across a 2x2 split for solo vs. relational motion and path vs. non-path motion. Examples within each split included solo {*idle*, *flinch*}, relational {*chase*, *flee*}, solo-path {*zigzag*, *spiral*}, and relational-path {*leapfrog*, *orbit*} verbs. For gesture, we chose 12 balanced across head {*nod*, *shake head*, *perk up*}, hand {*point*, *clap*, *wave*}, and body {*shrug*, *sag*, *dance*} gestures. Lastly for emotions, we chose 12 emotions balanced for valence: {*ecstatic*, *happy*, *surprise*, *awe*, *relaxed*, *tenderness*, *anger*, *confusion*, *disgust*, *fear*, *boredom*, *sadness*}. The full balanced list of stimulus words and their category splits can be found in the Supplementary Material.

For a four-alternative forced-choice setup (4AFC), we needed to generate fair distractors to be the other multiple-choice answers. We did this by using an embedding-based approach. First, we computed embeddings for each word using all-MiniLM-L6-v2 from



**Figure 3: fog let users directly manipulate speed, intensity: A) fast, erratic, exaggerated motion. B) slow and subtle. C) User-drawn path. E-H) Refinement panel dynamically generated minimaps with axes bespoke to the function composition.**

Sentence-BERT [21] and created a pairwise distance table. We selected the two closest distractors from within-category and one distractor from outside the category. For example, if we were evaluating an emotion like *ecstatic*, *happy* and *surprise* could be within-category (positive valence) distractors while *fear* would be an out-category distractor (negative valence). We then generated a fog implementation for each stimulus word in batch with factory generation scripts for each class. Each class implementation was done with Claude-Opus-4-6, with maxTokens at 12,000 tokens and a thinking budget of 10,000 tokens. The prompt was the abstract class definitions and minimal guidelines (e.g. no imports, speed ranges to anchor it in what values were fast or slow). We did not pass in few-shot examples. Our prompt pipelines are detailed in Supplementary Material.

Next, we conducted our crowdworking study on CloudResearch Connect. On a webpage showing four animations, each crowdworker was asked to judge which of the four choices best fit the motion expressed within the animation. For example, if the circle was approaching another character fast, was the verb *follow*, *fight*, *avoid*, or *cling*? If the circle was moving slowly towards the bottom of the screen and shrinking slightly in scale, was the emotion *fear*, *sadness*, *anger*, or *surprise*? This 4AFC setup meant our baseline was chance (guessing) at 25%. Each crowdworker rated all the stimulus words for the dimension, meaning they went through all 12 of the adverbs, gestures, and emotions, or all 32 of the verbs. The median time for this experiment ranged from 2.30 min (Gestures) to 4.25 min (Verbs), including the time taken to do two practice trials. We did a posthoc inspection of crowdworkers to flag for 1) outlier time taken 2) low accuracy below chance, and 3) position bias (always clicking the same answer). For each stimulus word, we had 10 people judge the animation and make a choice. Participants were paid \$0.75 for a 3 min task, which maps to a \$15/hr wage.

**5.1.2 Perceptual Evaluation Results.** We show the exact match accuracy results for all categories and compositions in Table 2. In aggregate, the overall accuracy for fog was 67.8%.

**Table 2: Recognition of Motion Function Composition in 4 Alternatives Forced Choice Task**

| Dimension      | N (stimuli x judgments) | Accuracy (%) |
|----------------|-------------------------|--------------|
| Verbs          | 320 (32x10)             | 77.7         |
| Adverbs        | 120 (12x10)             | 62.5         |
| Gesture        | 120 (12x10)             | 78.0         |
| Emotion        | 120 (12x10)             | 64.2         |
| Adverb + Verb  | 192 (8x8x3)             | 58.3         |
| Gesture + Verb | 192 (8x8x3)             | 69.8         |
| Emotion + Verb | 192 (8x8x3)             | 57.8         |
| Overall        | 1256 (260 stimuli)      | 67.8         |

All accuracy was significant ( $p \leq 0.01$ ) compared to a chance baseline (25%).

**5.1.3 Verbs.** Verbs were identified with a decently high exact match accuracy at 77.7% across 32 verbs. Aggregating across all judgments, the most recognizable verbs were: {*glide*, *zigzag*, *wander*, *spiral*, *flee*, *struggle*, *bounce*, *orbit*} (10/10 agreement), *chase*, *avoid*, *repel*, *guard*, *lunge*, *idle*, *watch* (9/10 agreement). While some of the best performing verbs did have recognizable paths, in aggregate, path verbs were accurate to 74.3%, and non-path verbs were accurate to 80.6%. Solo verbs tended to do better (82.2%) than relational verbs (72.2%). This could have been because the other entity’s motion was controlled and not actively responsive to the tested verb. The worst performing verbs ( $\leq 2/10$ ) were *intercept*, *mirror* and *escort*; the latter two were often caught in confusion pairs with *cling*.

**5.1.4 Adverbs.** Adverbs were shown with *walk* as the base verb, so users could have an animation to anchor the modified motion on. The exact accuracy for adverbs was 62.5%. Aggregating across all raters, the most recognizable adverbs were *slowly* and *quickly* (10/10), *hesitantly* (9/10), *playfully* (8/10). The worst performing adverbs were *forcefully* (0/10) and *erratically* (4/10). One reason for the modest accuracy is that many of the worst performers were often caught in confusion pairs: *forcefully*-*erratically* (7/10 times), *erratically*-*playfully* (6/10), and *suddenly*-*quickly* (5/10).

**5.1.5 Gesture.** The exact accuracy on gestures was 78.0%. By category, head gestures had exact match accuracy of 80%, hand gestures 95%, and body gestures 68%. The most recognizable gestures were stretch, wave, dance, hands-up (11/11) and point, clap, shake head (10/11) and nod (9/11). The least recognizable gestures were tilt (5/11) and shrug (1/11). Shrug was commonly confused with other gestures like *stretch* or *clap*, which the implementation shared motion components with (stretching, hand movement).

**5.1.6 Emotions.** The exact match accuracy on emotions was 64.2%. The top emotions recognized were sadness (9/10), boredom (9/10), anger (9/10), ecstatic (8/10), confusion (8/10), fear (7/10), awe (7/10). Negative emotions were easier to get right (77%) than positive emotions (52%). The most common confusions in emotions were *happy* vs. *ecstatic* (5/10), *surprise* vs. *ecstatic* (4/10), *relaxed* vs. *tenderness* (4/10).

**5.1.7 Function Composition Evaluation Methodology.** To also evaluate function compositions, we designed a second set of evaluations to look at Adverb(Verb), Emotion(Verb), Verb+Gesture. From each dimension, we took the top 8 most recognizable stimulus words from each dimension and crossed them to create 64 (8x8) function compositions. We wanted to make sure that each function composition was evaluated by at least 3 participants. However, each participant had to see only a subset of the full compositions, because otherwise they could learn the motion signature of certain stimulus by coming across repeats. To mitigate for such learning effects, we used a Latin-Square setup. This meant participants each stimulus word only once during their trials. During the perceptual evaluation, participants again used a webpage that gave them an introduction and two practice trials with stimulus words that were not in the eval set. On average, participants took 4 minutes with the task and were paid \$0.75 for the task (same as before).

**5.1.8 Adverb + Verb.** The Adverb(Verb) composition accuracy was 58.3%, with the adverb component being recognized 70.3% of the time and the verb component being recognized 85.9% of the time. Again, some of the most common confusions were due to adverb confusion pairs like *urgently*-*eagerly* and *lazily*-*hesitantly*. One reason why adverbs and verbs sometimes did not mesh well in compositions, was because their implementations either targeted different channels or overwrote each other. For example, the set of adverbs {*hesitantly*, *eagerly*, and *nervously*} all targeted acceleration, but certain verbs had behavior that was driven mainly by speed, exaggeration, and easing rather than acceleration, so there was channel misalignment.

**5.1.9 Verb + Gesture.** The Verb+Gesture composition accuracy was 69.8%, with the verb component recognized at 87.0% accuracy and gesture component recognized at 78.6% accuracy. Examples of best performing compositions included *orbit*+*wave*, *flee*+*hands up*. Gestures often targeted the motion on hand particles, leaving the entity freer to express a path or primary motion. However, the function composition could also have both functions writing to the same channel and still be recognizable. For example, *wander*+*shake head* both wrote primarily to velocity and squash-and-stretch but the temporal dynamics of the head shake (two back-and-forth motions) made it distinguishable from *wander*.

**5.1.10 Emotion + Verb.** The Verb+Emotion composition accuracy was 57.8%, with the verb component recognized at 72.4% accuracy and gesture component recognized at 80.7% accuracy. Easily recognized compositions included *ecstatic*+*spiral*, *sadness*+*struggle*, and *boredom*+*wander*. Compositions that were not recognizable included *happy*+*struggle* and *sadness*+*chase*. The low accuracy on these latter compositions make sense considering they do not immediately connect as emotion-verb pairings.

**5.1.11 Function Composition Self-Refinement.** We carried out one last experiment to look at function composition self-refinement. We provided an LLM with the implemented classes and told it to refine the secondary motion specifically for the primary motion (e.g. refine aggressively specifically for verb|chase). The prompt instruction was to adapt the decorate method and resolve channel mismatches to better share state. Prompt details for self-refining function compositions can be found in Supplementary. The

methodology is one-to-one with the original function composition methodology (Latin-Squares, three judgments per rater).

We did not see significant improvement in overall accuracy. For refined AdverbVerbs, the exact match accuracy was 58.9% (+0.6%). For refined GestureVerbs, the exact match accuracy was 64.6% (-5.2%). For refined EmotionVerbs, the exact match accuracy was 60.4% (+2.6%). The component accuracies fluctuated both above and below the original function composition component accuracies (see Supplementary Material).

**5.1.12 Motion Recognition Perceptual Evaluation Conclusion.** These experiments showed support for H1 – *fog* could express a wide range of motion signatures across all dimensions (67.8% overall accuracy, up to 78.0% on dimensions like verb and gesture). However, analysis showed that function composition could be flawed if the channels read and written to did not align. A self-refinement phase was tested, but it did not appear to offer significant improvements when applied automatically.

## 6 User Study

Next, we conducted a user study to centered around the following research questions:

*RQ1) To what extent can fog support animation editing in an interactive context?* Our technical evaluation showed that there will always be a margin of error as functions are generated on-the-fly. We wanted to understand how users would react and adapt to this compared to the standard prompt paradigm.

*RQ2) To what extent does fog's suite of interaction techniques (decorators, direct manipulation, and function composition refinement UI) support animation iteration?* Our hypothesis was that *fog* would alleviate some problems that come from AI execution and prompt interaction, but that the function composition framework could also introduce some new friction from extra layers of indirection.

### 6.1 Methodology.

We recruited participants on Discord and Upwork by posting for people interested in AI, animation, and motion graphics. A screening survey helped filter participants based on expertise and interest in AI and animation. This allowed us to find a balanced set of six professionals and four novices. We decided to include a mix of expertise, because we felt that the system would benefit from having perspectives from professional disciplines like animation, coding, and game design. All participants were paid \$25 for an hour of their time. Prior to the study, participants were sent an information sheet and consent form. This methodology is covered by a relevant IRB protocol.

**6.1.1 Baseline.** We constructed a baseline that was prompt-only by creating a wrapper over an LLM. This LLM was Claude-Opus-4-6, which had 16,384 maximum tokens with a thinking budget of 3,000 tokens per prompt. In this baseline version, prompts returned an animation state configuration with raw code for each state. The code did not have to adhere to *fog*'s framework. This was a fair baseline because the LLM could write its own helper functions to fit the prompt with less indirection than *fog*. For visual parity, the entity still had velocity-dependent squash-and-stretch and a motion

trail. This ablated version also included a timeline representation so users could understand how to refer to states in the animation.

During the user study, participants were given an introduction to *fog*'s interface. The experimenter gave them a tutorial walkthrough and allowed them to warm up with each of its features. They were then told that their experimental task was to animate two simple stories with shapes. They would start editing from a base animation that either showed a prosocial (teaching, friendship) or antisocial scenario (e.g. rejection, taunting). They had a maximum of 20 minutes to edit the animation within each system. To minimize learning effects, we randomized the order of the prototype and scenario type across participants. Between each task, we had participants fill out a survey with Likert-scale questions. Additionally, we instrumented the prototype to log interaction traces for analysis.

### 6.2 User Study Results

Overall, participants described both versions of *fog* to be enjoyable and performant experiences. Novices and professionals alike could come up with creative concepts to express emotion and social dynamics. They tried concepts like “enemies to lovers”, “cat and mouse”, “a character coming out of a panic attack”, “friends playing outside while mom is angry at home” – all complex storylines and dynamics acted out in simple shapes. We first report quantitative metrics observed across all participants, before reporting qualitative feedback from professionals and novices.

**6.2.1 Efficiency, Latency, and Feedback Cycles.** On average, participants were about three times more efficient in taking editing actions with the *fog* interface than with the baseline. In the baseline, the generation actions were the prompts, and on average participants put in 5.2 prompts (min=2, max=9) before stopping. In *fog*, editing actions included function generation, global prompts, and function composition refinements, draw actions, and XY minimap exploration. We excluded selection actions when users tested presets and deduped direct manipulation events. On average, participants took an average of 15.1 (min=2, max=30) generation actions in *fog*. A paired t-test validated that this 3x improvement was significant ( $p \leq 0.01$ ).

Next, we looked at the time in between new animations to get a measure of how fast the feedback cycle was for users in both tasks. We derived this from user logs, summing the number of unique animations participants saw and dividing by the time taken for tasks. In the baseline, participants spent on average 1.75 minutes between new animations, waiting depending upon the complexity of their edit and animation scene, with a wait time that tended to grow as the task went on. Participants could get very verbose and also vague, and LLM take longer to think and return. In *fog*, the time between new animations was 0.36 minutes between new animations. A paired t-test on per-participant rates (unique animations / time taken) for *fog* and the baseline showed that this difference was significant ( $p \leq 0.001$ ). Many participants also verbally acknowledged the more rapid feedback cycle and sense of lower latency as they explored function compositions, presets, paths, and parameters in parallel with generation.

**6.2.2 Motion Refinement.** Our hypothesis for RQ2 was that *fog* could alleviate certain frictions (help people directly manipulate

speed, path, and positioning rather than trying to describe the edit). However, we also thought that imposing a function composition grammar could introduce new friction and indirection – users and AI had to work within the bounds of the grammar. Our survey metrics were aligned with this hypothesis. Most participants rated fog positively for Exploration (9/10) and Ability (7/10), whereas Control ratings were mixed (4/10). We provide context to these survey metrics with qualitative findings, which describes when fog provided more precision, exploration, and iterability and when users faced friction.

**6.2.3 Function Generation for Open-Ended Motion Vocab.** Participants liked the ability to suggest and generate new motion functions. They were able to successfully generate new verbs such as {*attack*, *trail*, *drawHome*}, new gestures such as {*beckon*, *slap*, *deflate*}, new adverbs such as {*lovingly*, *smoothly*} (adverbs), and new emotions such as {*volatile*, *insane*, *bitterDread*}. They liked to experiment with motion functions even when they were out of the bandwidth of what the system could explicitly express (circles could not smile). Examples of function compositions in scenes included aggressively(chase)+slap, where one shape would slap another shape upon arrival, or beckon on one shape and follow on another – the interaction created the look of beckoning. As an example of participant editing, P10 started from a base animation where a triangle approached the smaller circle of two circles. They edited it to look more like a kidnapping scene. They used *repel*, *fight*, *urgently*, *trail*, *anger* to create a constant fighting and colliding interaction between the bigger circle (guardian) and triangle (aggressor) as the triangle took the smaller circle (child). Other participants (P5, P8) also independently tried to create fight scenes with the baseline but often would get stuck on small corrections (e.g. "make them touch / collide / not overlap like that") and lose focus on the motion dynamics. However, participants still tended to rate the baseline positively for control, because while it could error, with a state-of-the-art foundation model at the backend, it could still meet most of their complex edits, and most participants did not try very precise edits with the baseline.

Participants liked the novelty of the function composition refinement panel. They explored emotion-emotion function compositions and emotion-verb compositions. For example, P8 generated the emotion *Volatile* by crossing *Anger* with *Stress*. The refinement panel returned a bespoke XY graph that had "volatility and intensity" as the tunable axes. When they moved around different pockets of the parameter space, they noticed that manipulating the "volatility" affected the speed and energy of the motion. "It's really nice how everything is connected. Changing the emotion changes the movement." Other participants also liked seeing bespoke axes like agitation versus intensity or reluctance vs. lethargy generate for their compositions. However, sometimes participants felt like the XY graph overpromised on control, because the delta in the animations as they explored the minimap could be too subtle.

**6.2.4 Trade-offs for Control.** Multiple participants said they preferred fog for its control (P7, P8, P9) and the ability to fine-tune towards more complexity (P10). Some participants like P9 preferred it purely for the ability to draw – having the entity follow their paths to them was fun and gratifying. However, other participants like P7, a novice, said it made it feel more like programming, like

one had to learn "cues". P8 felt like fog was better for "the artist or the perfectionist" because they felt the editing in it was more atomic and detailed. However, they also noted that in contrast to the baseline, fog could sometimes end up with function compositions where the effects cancelled out or subdued each other. Additionally, for some participants, while the baseline could always change the source code directly towards the prompt, fog could sometimes end up with a less satisfying result when it: 1) reached for bad combinations of available presets (aggressively(walks)), 2) came back with function compositions that composed too much at once (multiple function compositions on multiple characters), or 3) accidentally overwrote states participants did not mean to touch. These issues made participants penalize fog on control.

**6.2.5 Expert Feedback.** Lastly, we touch upon feedback from professionals and novices. Many professionals commented how they liked that fog put a special focus on emotion. E4, a junior animator, commented "I can see emotions in the circle. It was very interesting to see because it's just a simple shape, and how it can be angry, stressed, or chill – I think it's amazing." P2, a filmmaker, liked that there were interactions specifically for speed, positioning, and expression of emotion – those were their greatest painpoints for AI animation. Recently, they had begun creating AI character animation, and they found it hard to get consistent emotional mixes like surprise + deadpan, so they saw the utility. Other professionals like P5 from a game design background also thought the motion function generation made sense, because "people will want to make their own behaviors and have specific ways they want the shape to behave." P5 often used scriptable objects in their development work to create enemy behaviors like *idle* and *patrol*. The Verb class was an extension of that. Adverbs and the way they created semantic wrappers over math was newer to them, they usually only worked with speed in terms of vectors and implementation details. Overall, experts saw the most potential in the ability to generate functions bespoke to their needs and workflow.

**6.2.6 Novice Feedback.** Novices also enjoyed the system as well as its concept and could also quickly begin animating. They appreciated other features of fog more, like the suggestability of the motion grid. Often, they described how it was hard to come up with ideas in the baseline condition. They also liked that the presets created a sense of predictability. P5 said, "I can't really predict how it's going to happen when I enter the prompt, but with the presets I can imagine." Novices were able to accomplish the editing tasks of expressing emotion and social dynamics quickly. P1 animated a circle to be anxious, as if it was having a panic attack, and then calmed down. They thought the ability of the system to express the emotion was impressive and described seeing themselves in the entity's motion. "I feel maternal towards it." Another participant, P9, made two friends move along a path they drew. They added a third character to represent a mother entity, generated a function that made the mother trace out the shape of a house, and made the mother angry the other two did not come back to the house. "I can see the emotion in the circles, and I like that." Overall, novices gravitated to the features that were self-explanatory (presets, pencil), and found the motion vocabulary grid already supplied well-enough.

## 7 Discussion

We introduced *fog* and its abstract class-based architecture for motion and emotion. We evaluated the range and interaction support of function composition in both automatic and interactive settings and now discuss how our *fog*'s approach and results can generalize to other problems related to code generation, motion, and world models.

**7.0.1 Class abstractions as a new prompt surface for AI.** Prompts are the de facto interaction paradigm for engaging with AI, but they are fundamentally dependent upon instruction following natural language. One of *fog*'s ideas is to move towards abstract class and interface definitions rather than prompts. *fog* was at first inspired by the definition of interfaces in Python. Interfaces as code representations define contracts for what behavior should be inherited and implemented. *fog* is light in classes and methods to implement. It is a simple architecture where function compositions work over a few contract channels (speed, exaggeration, decoration, particles). Other class architectures could be imagined for more specific problems like the animation of assemblies and robots, or the choreography of movement in virtual space. Furthermore, *fog* showed how to benchmark this class inheritance approach with automatic evaluation and find the margin of error with a low number of evaluators doing light work. This sort bespoke benchmarking can be helpful as people move towards generative user interfaces, where features, functions, and UI generated on-the-fly will be more common. Users will still expect to have some traditional or built-in features, however, and we found in our user evaluation that *fog* could struggle at the choice between picking from the built-in vocabulary and generating a new function – so this represents a starting point for future work.

**7.0.2 Physics versus choreography.** In its early concepts, *fog* focused on physics-grounded animation: force-driven composition, event-driven states, and stateful functions. However, we found that getting the physics alone was not enough to express social dynamics – characters don't perfectly transfer energy or downramp velocity. Characters move intentionally *against* physics and gain energy from internal motivation like emotions and goals. Getting a bounce that followed the principles of gravity right was not as effective as getting squash-and-stretch and exaggeration right for this system. Pure functions (like the ones for easing and interpolation-driven path motion) were better than stateful functions accumulating force and state. Once just a few core physics were in place, *fog* was better off focusing on the choreography of the motion, how entities were staged and timed and expressed relative to each other, through coordination and opposition. *fog* happened to be designed in this way, based on opinion and exploration, but other architectures and different ratios of support between physics and choreography could have worked just as well. In generative animation and video, does it really matter that an apple falls with a gravity of 9.8 meters per second squared? Perhaps these questions will be left to user choice (storytelling or simulation?), and system design will come to users.

**7.0.3 Generalizability, Future Work, Limitations.** The most obvious limitation to our work is that there are limits to what one can express in Heider-Simmel animations. Circle- and triangle-land is not Hollywood. While our accuracy was 68%, and there could

be room for improvement, we could not expect 100% or even 90% either – animating a circle cannot express all of the possible verbs, adverbs, and emotions out there. Participants also felt this; often, *fog*'s suggest feature return suggestions like flutter or shadow, things the system was inequipped to express on a shape without lighting and more parts to move. This led participants to feel like the system could overpromise in suggestion and underdeliver in implementation in comparison to the baseline.

Additionally, there are also probably limits to the code representations of motion and emotion in general, and we think there could be other valuable continuous representations to explore. Code most often represents animation in terms of position, rotation, scaling, paths, and randomness. However, there are other representations like performance input that could easily get us more realistic motion dynamics and break us out of the box of these limited degrees of freedom. For example, a user could perform a motion and retarget it from motion capture or video. With a small set of performances, it would be interesting to see if one could teach AI through reinforcement learning or simple ML architectures to create new behaviors from demonstration. Another separate, interesting, and immediate idea for future work could be to see if these abstract shape animations can be used as motion prompts for video models.

## 8 Conclusion

*fog* is a function composition framework of abstract classes that bridge motion and emotion. It introduces methods to create an open-ended vocabulary of motion spanning Verbs, Adverbs, Gestures, and Emotions. To demonstrate the expressive range of *fog*, we conducted a motion recognition perceptual evaluation with 452 animations, finding that people can recognize the semantic meaning of the motion over 68% of the time. To demonstrate its capabilities for storytelling and user control, we conducted a mixed-methods evaluation with professionals and novices, who animated Heider-Simmel stories to express emotion and social dynamics. We found that *fog* could support editing with rapid iteration and structured exploration.

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